Assignment ครั้งที่ 7

วิชา 060243104 Object-oriented Programming

ตอนที่ 3

โดย

68-060216-1203-7 นายปภาวิน ธิติชุณหกุล

ห้อง IT 1-RC

เสนอ

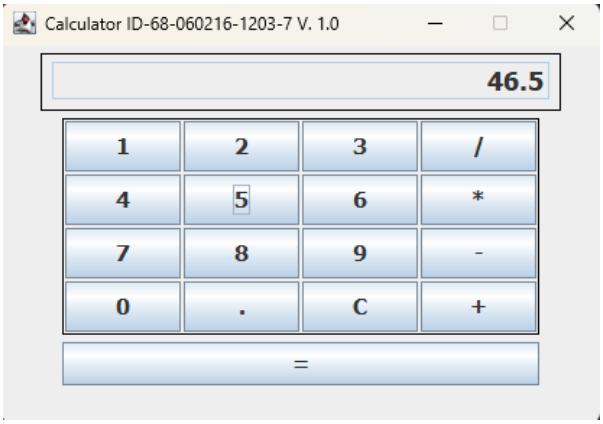
ผู้ช่วยศาสตราจารย์สมชัย เชียงพงศ์พันธุ์

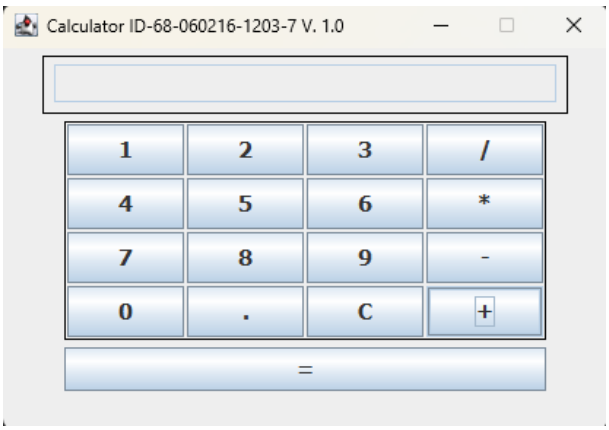
ภาควิชาเทคโนโลยีสารสนเทศ

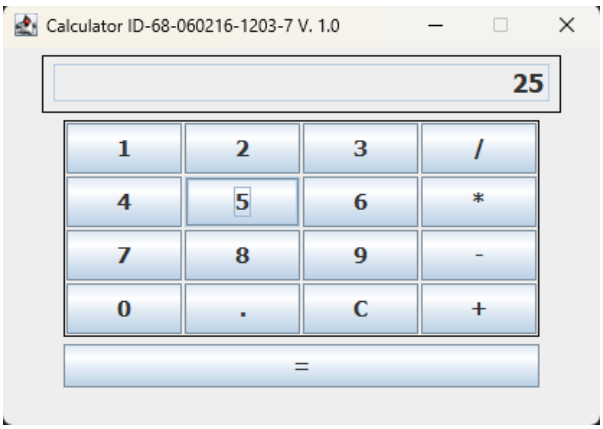
คณะเทคโนโลยีและการจัดการอุตสาหกรรม

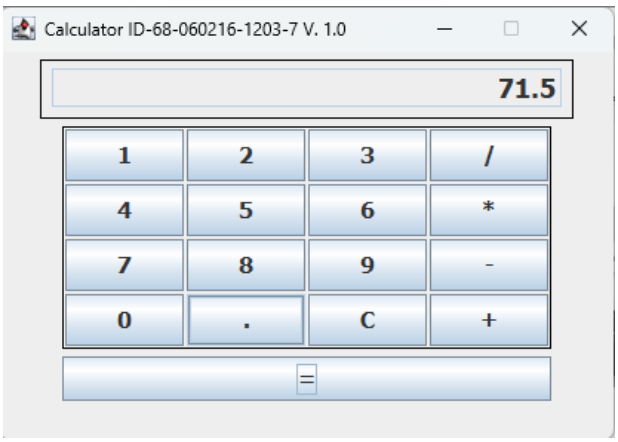
Assignment #07
060243104 Object-oriented Programming

```java
// Id    :  68-060216-1203-7
// Name  :  Mr.Paphavin Thitichunhakun
// Room  :  1 RC
// File Name Calculator_6806021612037.java

import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JPanel;
import javax.swing.JTextField;
import javax.swing.border.LineBorder;

import java.awt.Color;
import java.awt.Container;
import java.awt.Dimension;
import java.awt.FlowLayout;
import java.awt.Font;
import java.awt.GridLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class Calculator_6806021612037 extends JFrame implements ActionListener  {
    JPanel panelResult, panelBtn, panelCal;
    JTextField textResult;
    JButton btn[];
    Container c = getContentPane();
    Font fontResult = new Font("Tahoma",Font.BOLD,18), font = new Font("Tahoma",Font.BOLD,15);
    String strList[] = {"1", "2", "3", "/", "4", "5", "6", "*", "7", "8", "9", "-", "0", ".", "C", "+"}, str = "", cal = "";
    Boolean Cal = false;
    double num1, num2, total;

    public static void main(String[] args) throws Exception {
        new Calculator_6806021612037();
    }

    @Override
    public void actionPerformed(ActionEvent event) {
        if (event.getSource() == btn[0]) { str += btn[0].getText(); textResult.setText(str);}
        else if (event.getSource() == btn[1]) { str += btn[1].getText(); textResult.setText(str);}
        else if (event.getSource() == btn[2]) { str += btn[2].getText(); textResult.setText(str);}
        else if (event.getSource() == btn[3]) {
            try {num1 = Double.parseDouble(str);}
            catch (Exception e) {}
            cal = btn[3].getText();
            str = ""; textResult.setText(str);
            Cal = true;
        }
        else if (event.getSource() == btn[4]) { str += btn[4].getText(); textResult.setText(str);}
        else if (event.getSource() == btn[5]) { str += btn[5].getText(); textResult.setText(str);}
        else if (event.getSource() == btn[6]) { str += btn[6].getText(); textResult.setText(str);}
        else if (event.getSource() == btn[7]) {
            try {num1 = Double.parseDouble(str);}
            catch (Exception e) {}
            cal = btn[7].getText();
            str = ""; textResult.setText(str);
            Cal = true;
        }
        else if (event.getSource() == btn[8]) { str += btn[8].getText(); textResult.setText(str);}
        else if (event.getSource() == btn[9]) { str += btn[9].getText(); textResult.setText(str);}
        else if (event.getSource() == btn[10]) { str += btn[10].getText(); textResult.setText(str);}
        else if (event.getSource() == btn[11]) {
            try {num1 = Double.parseDouble(str);}
            catch (Exception e) {}
            cal = btn[11].getText();
            str = ""; textResult.setText(str);
            Cal = true;
        }
        else if (event.getSource() == btn[12]) { str += btn[12].getText(); textResult.setText(str);}
        else if (event.getSource() == btn[13]) {
            Boolean dod = false;
            for (int i = 0; i < str.length(); i++) if (str.charAt(i) == '.') dod = true;
            if (!dod) str += btn[13].getText(); textResult.setText(str);
        }
        else if (event.getSource() == btn[14]) { num1 = 0; num2 = 0; total = 0; str = ""; textResult.setText(str);}
        else if (event.getSource() == btn[15]) {
            try {num1 = Double.parseDouble(str);}
            catch (Exception e) {}
            cal = btn[15].getText();
            str = ""; textResult.setText(str);
            Cal = true;
        }
        else if (event.getSource() == btn[16]) {
            if (Cal == true) {
                try {num2 = Double.valueOf(str);}
                catch (Exception e) {}
                str = ""; textResult.setText(str);
                if (cal.equals("/")) textResult.setText(String.valueOf(total = num1 / num2));
                else if (cal.equals("*")) textResult.setText(String.valueOf(total = num1 * num2));
                else if (cal.equals("-")) textResult.setText(String.valueOf(total = num1 - num2));
                else if (cal.equals("+")) textResult.setText(String.valueOf(total = num1 + num2));
                num1 = total;
                Cal = false;
            }
            else {}
        }
    }

    public Calculator_6806021612037() {
        super("Calculator ID-68-060216-1203-7 V. 1.0");
        c.setLayout(new FlowLayout(FlowLayout.CENTER));
        initGui();
        setSize(430,300);
        setVisible(true);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setResizable(false);
    }

    public void initGui() {
        initPanelResult();
        initPanelBtn();
        initPanelCal();
    }

    public void initPanelResult() {
        panelResult = new JPanel(new FlowLayout(FlowLayout.CENTER));
        panelResult.setPreferredSize(new Dimension(360, 40));
        panelResult.setBorder(new LineBorder(Color.black));
        panelResult.add(textResult = new JTextField(20));
        textResult.setFont(fontResult); textResult.setHorizontalAlignment(JTextField.RIGHT);
        textResult.setEditable(false);
        c.add(panelResult);
    }

    public void initPanelBtn() {
        panelBtn = new JPanel(new GridLayout(4, 4, 2, 2));
        panelBtn.setPreferredSize(new Dimension(330, 150));
        panelBtn.setBorder(new LineBorder(Color.BLACK));
        btn = new JButton[strList.length + 1];
        for (int i = 0; i < strList.length; i++) {
            btn[i] = new JButton(strList[i]);
            btn[i].setFont(font);
            btn[i].addActionListener(this);
            panelBtn.add(btn[i]);
        }
        c.add(panelBtn);
    }

    public void initPanelCal() {
        panelCal = new JPanel(new GridLayout(1,1));
        panelCal.setPreferredSize(new Dimension(330, 30));
        btn[strList.length] = new JButton("="); btn[strList.length].setFont(Font);
        btn[strList.length].addActionListener(this);
        panelCal.add(btn[strList.length]);
        c.add(panelCal);
    }


}

```

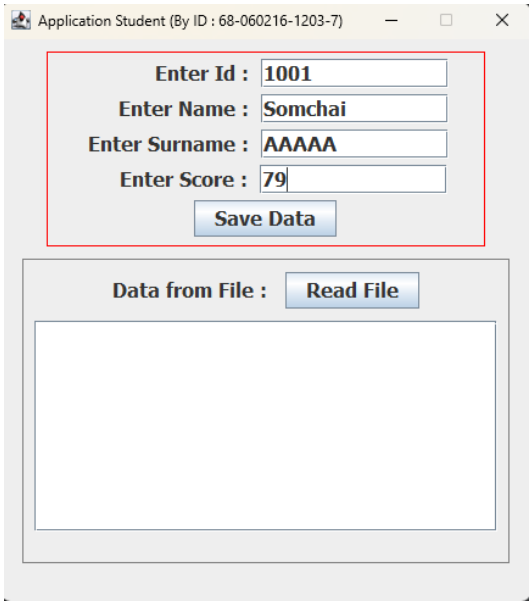

Calculator ID-68-060216-1203-7 V. 1.0
46.5
1 2 3 /
4 5 6 *
7 8 9 -
0 . C +
=

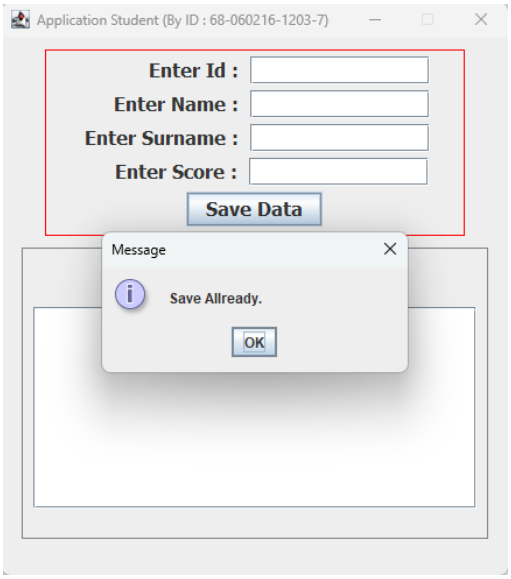

Calculator ID-68-060216-1203-7 V. 1.0
1 2 3 /
4 5 6 *
7 8 9 -
0 . C +
=

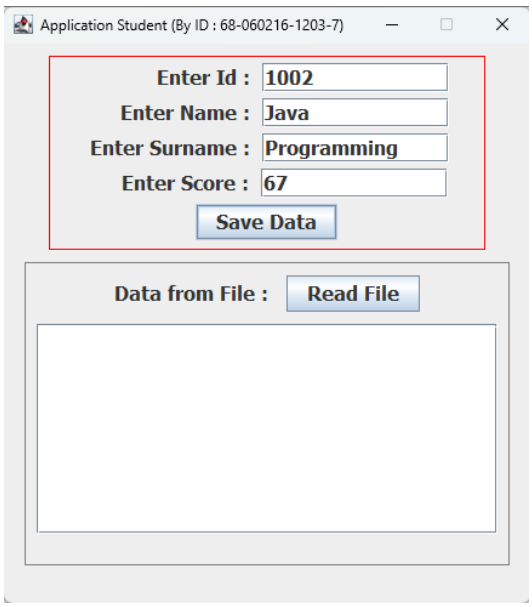

Calculator ID-68-060216-1203-7 V. 1.0
25
1 2 3 /
4 5 6 *
7 8 9 -
0 . C +
=

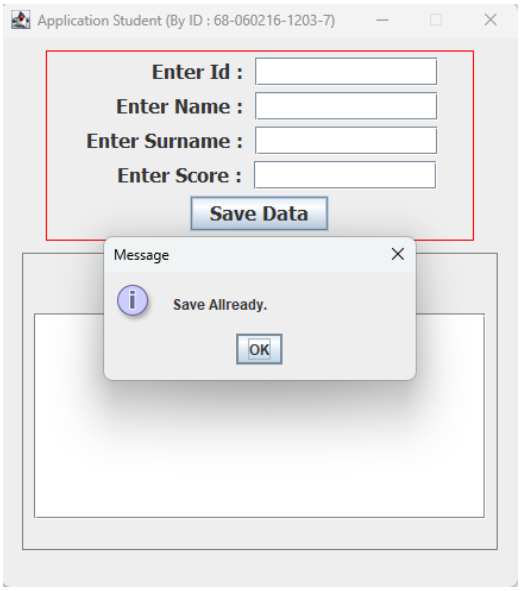

Calculator ID-68-060216-1203-7 V. 1.0
71.5
1 2 3 /
4 5 6 *
7 8 9 -
0 . C +
=

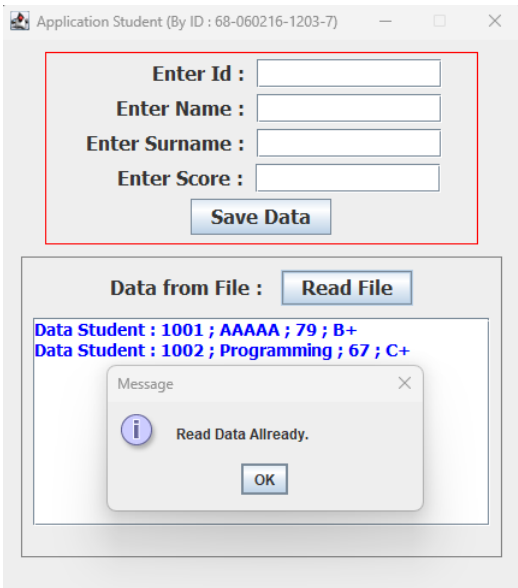
```java
// Id    :  68-0602IG-1203-7
// Name  :  Mr.Paphavin Thitichunakan
// Room  :  1 RE
// File Name Assign07_6806021612037.java

import java.awt.Color;
import java.awt.Container;
import java.awt.Dimension;
import java.awt.FlowLayout;
import java.awt.Font;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.FileReader;
import java.io.FileWriter;

import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JOptionPane;
import javax.swing.JPanel;
import javax.swing.JScrollPane;
import javax.swing.JTextArea;
import javax.swing.JTextField;
import javax.swing.border.LineBorder;

public class Assign07_6806021612037{
    public static void main(String[] args) {
        new AppStudentGrade();
    }
}

class AppStudentGrade extends JFrame implements ActionListener {
    JPanel panelEnter, panelResult;
    JTextField textId, textName, textSurname, textScore;
    JButton btnSave, btnRead;
    JTextArea textArea;
    JScrollPane scrollPane;
    Container c = getContentPane();
    Font font = new Font("Tahoma", Font.BOLD, 16);
    Student std;

    public AppStudentGrade() {
        super("Application Student (By ID : 68-0602IG-1203-7)");
        initGui();
        setSize(450, 500);
        setVisible(true);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setResizable(false);
    }

    @Override
    public void actionPerformed(ActionEvent event) {
        if (event.getSource() == btnSave) {
            try {
                std = new Student(textId.getText(), textId.getText(), textSurname.getText(), Integer.parseInt(textScore.getText()));
                BufferedWriter writer = new BufferedWriter(new FileWriter("Students.txt", true));
                writer.write(std.toString() + "\n");
                writer.close();
                textId.setText(""); textName.setText(""); textSurname.setText(""); textScore.setText("");
                JOptionPane.showMessageDialog(c, "Save Already.");

            } catch (Exception e) {JOptionPane.showMessageDialog(c, e, "Error Message", JOptionPane.ERROR_MESSAGE);}
        }
        else if (event.getSource() == btnRead) {
            try {
                String output = "";
                BufferedReader reader = new BufferedReader(new FileReader("Students.txt"));
                for (String str : reader.readAllLines()) output += (str + "\n");
                textArea.setText(output);
                reader.close();
                JOptionPane.showMessageDialog(c, "Read Data Already.");
            } catch (Exception e) {JOptionPane.showMessageDialog(c, e, "Error Message", JOptionPane.ERROR_MESSAGE);}
        }
    }

    public void initGui() {
        c.setLayout(new FlowLayout(FlowLayout.CENTER, 10, 10));
        initPanelEnter();
        initPanelResult();
    }

    public void initPanelEnter() {
        panelEnter = new JPanel(new FlowLayout(FlowLayout.CENTER, 5, 5));
        panelEnter.setPreferredSize(new Dimension(360,160));
        panelEnter.setBorder(new LineBorder(Color.RED));
        panelEnter.add(createJLabel("          Enter Id : "));
        panelEnter.add(textId = new JTextField(10)); textId.setFont(font);
        panelEnter.add(createJLabel("      Enter Name : "));
        panelEnter.add(textName = new JTextField(10)); textName.setFont(font);
        panelEnter.add(createJLabel(" Enter Surname : "));
        panelEnter.add(textSurname = new JTextField(10)); textSurname.setFont(font);
        panelEnter.add(createJLabel("      Enter Score : "));
        panelEnter.add(textScore = new JTextField(10)); textScore.setFont(font);
        panelEnter.add(btnSave = new JButton("Save Data")); btnSave.setFont(font); btnSave.addActionListener(this);
        c.add(panelEnter);
    }

    public void initPanelResult() {
        panelEnter = new JPanel(new FlowLayout(FlowLayout.CENTER, 10, 10));
        panelEnter.setPreferredSize(new Dimension(400,250));
        panelEnter.setBorder(new LineBorder(Color.gray));
        panelEnter.add(createJLabel("Data from File : "));
        panelEnter.add(btnRead = new JButton("Read File")); btnRead.setFont(font); btnRead.addActionListener(this);
        panelEnter.add(scrollPane = new JScrollPane(textArea = new JTextArea(10,25))); textArea.setForeground(Color.BLUE); textArea.setFont(new Font("Tahoma", Font.BOLD, 14));
        c.add(panelEnter);

    }

    public JLabel createJLabel(String title) {
        JLabel label = new JLabel(title);
        label.setFont(font); return label;
    }
}

class Person {
    private String name, surname;

    public Person() {
    }

    public Person(String name, String surname) {
        this.name = name;
        this.surname = surname;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getSurname() {
        return surname;
    }

    public void setSurname(String surname) {
        this.surname = surname;
    }

    @Override
    public String toString() {
        return "" + getName() + " ; " + getSurname() + " ; ";
    }
}

class Student extends Person {
    private String id;
    private int score = 0;

    public Student() {
    }

    public Student( String id, String name, String surname, int score) {
        super(name, surname);
        this.id = id;
        this.score = score;
    }

    public String getId() {
        return id;
    }

    public void setId(String id) {
        this.id = id;
    }

    public int getScore() {
        return score;
    }

    public void setScore(int score) {
        this.score = score;
    }

    public String toGrade() {
        if (score >= 80) return "A";
        else if (score >= 75) return "B+";
        else if (score >= 70) return "B";
        else if (score >= 65) return "C+";
        else if (score >= 60) return "C";
        else if (score >= 55) return "D+";
        else if (score >= 50) return "D";
        else return "F";
    }

    @Override
    public String toString() {
        return "Data Student : " + super.toString() + getScore() + " ; " + toGrade();
    }
}
```


Application Student (By ID : 68-060216-1203-7)
Enter Id : 1001
Enter Name : Somchai
Enter Surname : AAAAA
Enter Score : 79
Save Data
Data from File :
Read File


Application Student (By ID : 68-060216-1203-7)
Enter Id :
Enter Name :
Enter Surname :
Enter Score :
Save Data
Message
Save Allready.
OK


Application Student (By ID : 68-060216-1203-7)
Enter Id : 1002
Enter Name : Java
Enter Surname : Programming
Enter Score : 67
Save Data
Data from File :
Read File


Application Student (By ID : 68-060216-1203-7)
Enter Id :
Enter Name :
Enter Surname :
Enter Score :
Save Data
Message
Save Allready.
OK


Application Student (By ID : 68-060216-1203-7)
Enter Id :
Enter Name :
Enter Surname :
Enter Score :
Save Data
Data from File :
Read File
Data Student : 1001 ; AAAAA ; 79 ; B+
Data Student : 1002 ; Programming ; 67 ; C+
Message
Read Data Allready.
OK